



THE INSTITUTE OF
MASTERS
of WINE

Theory Report

2023 Stage 2 MW Exam

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Theory Exam Chair Report, Beverley Blanning MW

One of the biggest challenges of the Master of Wine examination is its scope. To achieve a pass, candidates need to be able to demonstrate expertise across the broad spectrum of the global wine industry. While this will not be news to anyone reading this, the point to consider here is that since all MW candidates work in some area of this diverse industry one would perhaps expect to see more people passing one or two papers in the areas where they have the most direct experience. But this is not what we saw in 2023, the first large cohort of examinees post-COVID. The stark reality is that two-thirds of candidates who failed this year's exam failed all five papers. Those who fall into this camp may take some comfort in knowing that they are far from alone, but it does pose wider questions about the general preparedness of candidates to sit the examination.

The report below provides a wealth of information on the 2023 theory papers, with valuable commentary from each Paper Chair on pitfalls to avoid and strategies for success. A number of well-worn themes can be seen. As well as displaying accurate knowledge, we are always looking for candidates to show critical analysis: this is a requirement for a passing grade at MW level. Every year, the examiners design questions that will allow candidates to show their understanding by analysing and considering all angles of a question, and going beyond mere description or repetition of something they may have been told by a winemaker or academic. Examples are key, but should be used to support arguments and should always be relevant to the question. It is not uncommon to see candidates shoehorning a limited range of examples into all of their essays, regardless of their relevance. This is not a recipe for success.

To add to the specific guidance on answering questions, a few words of general advice. Remember, this is an examination that tests ability to communicate in writing, generally in the specific format of an essay. This is a skill that requires practice. Practice writing essays to time and under exam conditions (especially with autocorrect spelling turned off). Give them to anyone who will read them for feedback. Allow time for planning, always strive for clarity and try to allow time to re-read essays. Always answer the question. Be honest with yourself about your ability to communicate fluently in writing. If English is not your first language, do consider translation.

Hopefully, the information contained in this report will lead to the future success of many who did not get their desired result this year.

Paper one report 2023: Paper chair, Rhys Pender MW

General Comments

Some positives this year were that most candidates presented papers quite clearly with an introduction, expanding on those points in the body of their essays and then finishing with a conclusion. Candidates close to a passing grade were often just missing that bit of critical thinking that shows their own thoughts on the question topics and then coming down with a well-considered answer. There was obvious knowledge, but not enough insight. There is still a tendency for many candidates to just repeat what respected producers are doing as the answer, rather than showing the critical analysis that is required to pass.

As mentioned in previous years, many candidates seem to use the same examples from the same source. While knowledge gained at seminars and popular conferences is important and useful, it helps to show a bit more global breadth in seeking out other examples from the broad range of the wine world, from bulk to fine wine. Another key factor missing in many answers this year was the consideration of cost and plausibility. Remember to always consider growers outside of the high-revenue vineyard areas and how they might tackle the issues presented and what might be realistic for them.

1. Extreme heat and drought threaten many wine regions. What steps can a vineyard manager take to mitigate their effects?

This was much the more popular choice of the two Section A questions, but with mixed results. Examiners were looking for a solid definition of heat and drought and then a multifaceted range of approaches that vineyard managers could take.

Candidates needed to show a full understanding of the options and also consider the reality of implementing these steps. We were looking for understanding of how heat and drought affect the vine and for this to be linked to the different steps available. It was important to consider all elements of the wine business, not just fine wine regions. Many candidates did not consider the cost implications versus the price per ton of grapes in different regions of the world. Many students gave unrealistic options that would not be possible, financially viable or even legal in some regions. The papers that went close but ultimately did not achieve a passing mark usually omitted a key element or focused too much on just one or two areas.

2. To what extent is excessive precipitation during the growing season a threat in the vineyard? How can it be managed?

Few attempted this question and it was not well answered. Typically, key elements were missed which made it hard to give passing grades. There were weaknesses in understanding of viticulture, of threats such as mildew and excessive growth, and of how vines respond to excessive precipitation. The question wasn't about how wet regions manage viticulture but about excessive precipitation, which can happen in any region.

The examiners were looking to assess candidates' understanding of the risks of excessive water to vineyard structure, leaching of specific nutrients, and difficulty in carrying out vineyard operations. It was important for answers to show an understanding of the differing effects of precipitation at different times during the growing season, e.g. flowering compared to harvest, and the approaches to managing this.

3. A viticulturist has observed a decline in vineyard yield over the last five years. Discuss the possible causes and how the issue might be addressed.

Important here was to look at declines over a five-year period, not one-off events that may cause a smaller vintage. Many essays drifted off-topic into one-off events such as heat spikes, frost, flood, etc.

A lot of essays were just very top level without showing the depth of understanding necessary. Many students focused too much just on one area (e.g. nutrients or vine disease) without spending enough attention on other possible causes. Good essays addressed multiple possible causes such as pests, disease, nutrition, soil health, water, cover crops, rootstock/vine choice, climate change and vine age.

A number of candidates just listed examples of what people are doing to deal with yield declines without showing any insight into why. For MW-level essays, it is not enough simply to report examples. We are looking for original thought and insight. There were also a number of papers that didn't answer the question, but instead answered one that the student hoped would be there, often an almost pre-written sounding essay on yields.

4. What are the causes and effects of nitrogen deficiency in a vineyard? How can it be remedied?

This topic seemed to deter a lot of candidates with only 17 students attempting it. However, those that did obviously knew the topic well with a nearly two-thirds passing. Those that failed mostly didn't spend enough time considering the causes and effects and focused too much on remedies. It was necessary to get down deep into issues like soil pH, organic matter, leaching, vigour, crop loads, cover crops, competition, vines and rootstocks, and then discuss how all these might affect the fruit quality and quantity, vine growth and nitrogen's impact through to fermentation. The remedies section was generally answered well but it goes to show again how important it is to understand and consider the "why" before being able to show knowledge through insight, backed by global examples to prove the "what".

5. Evaluate the potential risks and rewards of choosing to plant ungrafted vines.

This was the second most popular question in section B and it proved to be tricky for some candidates. Still, more than a quarter passed. Those who passed did so with a careful intertwining of showing an understanding of viticulture, commercial realities of running a vineyard, technical details around rootstocks, soil type and scion compatibility, phylloxera, nematodes, replanting/layering/propagation, etc., with clear evaluation of the potential risks and rewards. Legal issues and marketing opportunities were also touched on by many. Some candidates, though, wrote more like a paper 5 essay, getting enamoured with the romance of old ungrafted vines and forgetting to answer the question asked.

Many candidates who were unsuccessful seemed to take it as a question solely about phylloxera or rootstocks, and focused too much on just these subjects, while many others just listed examples of producers with ungrafted vineyards, parroting unproven claims to quality without any evaluation of the potential risks and rewards demanded by the question. The question was worded “choosing to plant”, so essays devoted to old, ungrafted vines missed answering the question.

6. Assess the role of sunlight in determining grape quality.

Nearly a third of candidates were successful in this question. It proved to be a good test of which candidates could delve deep into showing an understanding of the link between sunlight and quality and those who could only give a very top level, basic regurgitation of a viticulture textbook without much insight.

The question referred to grape quality. Many essays focused not on grapes but either vine health, or, more commonly, wine styles. Candidates needed to show an understanding of how the natural factors (aspect, altitude, latitude, etc.), combined with vineyard management can be used to manipulate sunlight exposure in a way that enhances grape quality, which also needed to be clearly defined.

The components of the grape, in particular sugar, tannin, acidity, anthocyanins, flavour and aroma precursors and aroma compounds needed to be discussed, and the balance of these elements linked to grape quality.

Photosynthesis, vineyard issues that can reduce sunlight (pests, nutrition, etc.), too much sunlight (sunburn, over ripeness), timing during the growing season (vigour, future yields), canopy management, row spacing and orientation, trellising, etc. all needed to be considered, options and scenarios presented and clearly linked back to grape quality.

Paper two report 2023: Paper chair, Richard Kershaw MW

General Comments

The 2023 exam witnessed an increase in candidate numbers, shedding light on their diverse approaches and varying knowledge standards. Some candidates displayed their wealth of information, backed by pertinent examples, and responded to questions accurately and professionally. In contrast, others relied on scant data, occasionally spiced up with hyperbole, hoping to scrape by. This year, possibly from fewer field trips undertaken, several papers often shoehorned examples haphazardly into unrelated essay sections or regurgitated generic seminar content.

Historically, students grapple with two primary challenges: question approach and example relevance. Effective essay planning at the outset and a comprehensive review at the end can prevent scripts that start with excessive description but run out of time before reaching the core argument. It still surprises examiners how poor introductions, missing definitions, and vague parameters persist in many essays. An essay's essence typically demands an in-depth discussion of several key points. Presenting two or three arguments may secure some marks but often falls short of addressing the question comprehensively. While some displayed erudition, their responses felt incomplete, implying a limited grasp of the subject.

Candidates must delve into the requisite technical details when armed with multiple valid points. The best answers do not merely state what, where, and when but also explore how and why things are done. A common pitfall is mentioning the "what" and "who" but failing to elucidate the "how" and "why," resulting in superficial answers, generalizations, and storytelling that, at times, is reminiscent of lifestyle journalism.

Field trips offer opportunities for students to visit global wine regions and glean insights from industry experts. Crafting the right questions to elicit pertinent responses is crucial. Interviewees often have subjective stances, necessitating substantiation to bolster your point. For instance, accepting a winemaker's claim at face value, like standing on one's head for wild fermentation, requires reasoned argument and rigorous cross-referencing!

In recent years, travel limitations have constrained first-hand experiences. However, adept candidates leveraged pre-COVID knowledge to augment their answers with pertinent facts and examples. Additionally, many students showcased limited field experience by excessively referencing the same region or winery throughout their essays, missing the opportunity for a global perspective.

Each statement must be substantiated by relevant examples. An opposing view enriches the argument. Poor scripts often relied on examples as the argument itself. To give an example: "Winery X uses yeast" is too vague and needs to be expanded to "Winery X uses this yeast because it promotes this characteristic"; further, an opposing view could be "Winery Y doesn't use this yeast because it enables this unwanted characteristic". Equally, avoid: "Winery X likes using this yeast to promote this characteristic (the assumption being, therefore, it must work)".

When citing specific wineries, providing sufficient location information aids examiners' understanding. Again, as an example: "Winery X, Central Otago" is satisfactory, but the more esoteric "Winery X, Serra Gaúcha, Brazil" needs further clarification. But refrain from "Bob from Winery Y," which means little.

Ambiguous spelling mistakes and incorrect usage of standard terms, such as "Cerevisiae," should be rectified for clarity. Moreover, the decision of some candidates, whose first language is not English, to write in English affects their ability to convey some of their points effectively. Such candidates should consider the option of translating for clarity.

1. Outline the techniques a winemaker can use to influence the colour of red wine, considering at least three different styles of wine.

The question garnered widespread attempts, with various approaches available to address it, including exploring diverse wine styles or adopting a chronological winemaking process perspective. The critical factors were careful reading, essay scoping, and well-structured thoughts. Many essays lacked adequate introductions, clear definitions, and a clear outline of their intended approach, resulting in muddled scripts.

The question offered an opportunity to mention thin- and thick-skinned grape varieties, which some candidates did. However, none referenced non-acetylated anthocyanins present in Pinot Noir or Nebbiolo, which could have bolstered their arguments.

Some candidates clearly possessed knowledge but struggled to apply it effectively. Some merely listed techniques without comprehending why a winemaker might choose a particular practice or the desired outcomes. Candidates often failed to delve into anthocyanins, a crucial aspect of colour enhancement. Understanding the anthocyanin-tannin relationship could have added depth to their papers.

The main body of the answer should have covered colour extraction methods during maceration (including cold soak, physical cap management, saignée, extended maceration, carbonic maceration, flash détente, and thermovinification), the use of maceration enzymes, pressing techniques, oxygen management during fermentation for colour stability, and the role of oak (chips and barrels) in colour stabilization. Additionally, addressing pH's role in colour, methods for pH adjustment, the impact of SO₂ on colour, and oxidation's effects on colour were expected. Other relevant points could have included yeast's role in colour adsorption and blending different grape varieties with distinct colour profiles.

Many candidates overlooked fundamental techniques such as cold soaking and extended maceration. Some scripts omitted key aspects like acidity and blending, while a few ventured into irrelevant topics, like rosé winemaking.

Repetitive use of the same examples without substantiating the point was common, detracting from the overall quality of responses.

2. Explain the methods winemakers use to influence the colour of rosé wines?

This question required a concise explanation of winemaking methods that impact the colour of rosé wine. The term 'explain' implies a focused breakdown of relevant procedures rather than a broader discussion of pros and cons.

Colour is pivotal in rosé wine production, shaping consumer perception and product success. It can also influence, rightly or wrongly, how consumers perceive sweetness levels in rosé wines. A strong answer should have outlined winemaking techniques influencing rosé wine colour, including methods to produce various hues and maintain colour integrity. An introduction could have touched on the wide range of rosé styles and colours, from pale to deep, and their perceived sweetness.

In-depth responses explored the relationship between colour extraction, phenolic structure, and stability during maceration, fermentation, and post-fermentation stages. Topics like the impact of light exposure on clear-glass bottled rosé (although technically paper 3) and the effect of oak aging on colour could also have been mentioned.

The focus should not have been merely be on making different rosé wines but on understanding how colour is influenced and manipulated. Strong answers distinguished between hue (e.g., pink vs. orange) and colour intensity (dark or pale).

Weaker answers overly focussed on the pale Provence-style rosé trend. Few responses considered grape selection as a primary colour influencer.

Methods for influencing rosé colour include: grape variety selection, skin contact duration, and temperature control during maceration. Techniques such as direct pressing, saignée, and blending should have been explained. Quality answers differentiated between these methods, notably direct pressing and maceration versus saignée. These candidates explained the reasons behind these choices, while weaker responses merely described them without rationale. The blending of red and white wines was mentioned, but few explored why this is restricted, apart from sparkling and even fewer mentioned that leached tannins from the red component may result in a harsher textured rosé.

Ensuring colour integrity involves managing pH levels, oxidation control, additives, clarifying agents, filtration, and SO₂ additions. Fining was commonly mentioned as a rather blunt method to reduce a darker colour, but few proposed better alternatives. Temperature control, pH, SO₂ management's impact on hue and intensity, and colour preservation during winemaking and short to medium-term aging were often overlooked. Destemming's impact was not widely discussed, and colour loss from winemaking to bottling was seldom acknowledged.

3. When and why might inert gas be used in still winemaking?

At first sight, this seemed a relatively straightforward question. A definition of inert gas was required, and explanation of the characteristics and properties of the different inert gases, along with their principal uses in winemaking. Answers should also have covered the techniques and strategies involved in effectively applying inert gas.

The principal reason for using inert gas is to safeguard wine from oxygen exposure and microbial contamination. Inert gases prevent oxygen uptake by wine. Answers should have discussed the effectiveness of different inert gases applied during grape transportation and receipt, pressing, racking, sparging and purging regimens, and selecting the correct gas for a particular application. Candidates should have mentioned techniques such as using dry ice bricks to stir up tank lees to lower oxygen ingress and referred to their use in flotation and carbonic maceration.

Many candidates struggled with the concept of "inert" and could not identify an inert gas correctly. Both Nitrogen and Argon needed mentioning, as did Carbon Dioxide (technically, CO₂ is not an inert gas but is used as an inert gas within the winery). Further, students could not state the properties of each inert gas, making it difficult to describe why they were used in each process. A few students incorrectly stated that the highly reactive gases SO₂ and Ozone are inert. Given these challenges, many students followed a descriptive essay of winemaking with examples of where inert gases were used rather than showing the critical analysis required by the question.

Examiners noted that virtually all the scripts assumed Argon was simply out of reach for most wineries. If those students had understood dosage rates, etc., many would have realised that using gases such as Argon, admittedly in only a few wineries, has high effectiveness yet at low dosage rates. Argon is not "exorbitantly" expensive, as one student remarked, but it is about three times the price of CO₂ and Nitrogen, making it worth considering given its effectiveness. In the essays that delved more deeply into the gas properties, lighter and heavier was the more common terminology, but terminology such as density would have been more appropriate.

Despite some of the shortcomings in inert gas understanding, it was still possible to pass if students covered most of its uses/scenarios and techniques. Unfortunately, many students kept the essay narrow and confined to the usual common examples of blanketing of grapes, flushing tanks and tank ullages, bottling and carbonic maceration. Less commonly mentioned were flotation, adjusting wine dissolved gas levels, tank mixing, etc.

Whilst marks are not deducted for typos, some spelling and grammar made it challenging to understand several scripts. Candidates should try to allow time to re-read their essays before the end to ensure clarity.

4. Which laboratory analyses should a winemaker carry out prior to alcoholic fermentation, and why?

The question addresses the laboratory analyses that winemakers should conduct before alcoholic fermentation and their significance in ensuring successful fermentation, preventing spoilage, and achieving the desired wine style. Candidates should have discussed the range of available analyses and, if possible, how these are determined, encompassing activities from grape picking to clarification before fermentation.

Few students attempted this question, likely due to its technical nature. It required specifying laboratory analyses before alcoholic fermentation, knowledge of relevant legal limits, and examples from around the world.

The pass rate for the question was low. Some candidates lacked the necessary technical knowledge and provided brief, superficial responses. Others mentioned the tests and their purposes but lacked critical thinking or elaboration on how the results influence winemaking decisions. This resulted in several candidates listing analyses without explaining why they were performed and the insights gained from the results for winemaking decisions. However, candidates with winery experience or immersive field trips did display a better grasp of the topic than those relying solely on textbooks.

Answers should have included analyses such as sugar levels (including glucose and fructose analysis) to assess ripeness; pH analyses to ensure optimal pH for yeast growth; titratable acidity to determine acidity levels affecting wine taste; nutrient status using YAN (and potassium levels) to provide necessary nutrients for yeast and prevent fermentation issues; and sulphur dioxide to prevent oxidation and spontaneous fermentation during settling. Other potential analyses could have included juice turbidity, total dissolved solids, microbial analysis, phenolic analysis, malic acid and, if applicable, smoke taint measurements, copper, and iron.

Strong responses considered the cost and practicality of various analyses, providing concise, relevant examples and emphasising the importance of taking a representative sample. One candidate mentioned volatile acidity, but this would be unusual to take from a juice sample preferment.

While the question allows for demonstrating a deep understanding of different wine styles and how technical parameters vary between them, many candidates treated all wines generically, missing this opportunity. Minor issues emerged, such as the careful use of abbreviations (full name first, followed by the acronym in brackets), providing units when stating numbers (e.g., 100 g/hL or 100 mg/L), and providing specific figures when describing pH levels (e.g., pH of 3.8 is high for sparkling base but not for fortified Muscat).

5. Write concise notes on four of the following additives:

Yeast nutrients

Oenological tannins

Bentonite

Carbon

Dimethyl decarbonate

This question examined knowledge of additives used in winemaking. Answers should have covered the function, application, source, and impact of each additive on the quality of wine, including a discussion of how these additives fit into the winemaking process, related safety concerns, the impact on wine quality and flavour, and legal regulations surrounding their usage in different regions.

Reading the question is always essential and in this instance a few candidates wrote an answer for all five categories, thus wasting valuable time that could have been put to better use. Many candidates lacked accuracy or technical knowledge in some aspects of the question, and few incorporated accurate addition rates for the different ingredients and their units of measurement. Despite the frequency of such questions over the years, very few answers covered four subjects in sufficient technical detail. Better answers combined knowledge of the chemical properties of the ingredient, along with when, how, and why they are added.

Yeast nutrients required an answer that included its role in supporting yeast growth and fermentation to provide optimal flavour, its measurement through YAN, when and why it is added to the must or wine, and the different types of commonly used nutrients.

A definition was important for oenological tannins and the available types.

Bentonite use in the winery needed a robust explanation covering its usefulness as a fining agent in winemaking to remove proteins and reduce concentrations of specific compounds. A definition of the two main types, sodium bentonites and calcium bentonites, and the rehydration process was helpful. Candidates then needed to identify the two main functions in winemaking: clarification and removal of haze-forming (heat-unstable) proteins.

Details about its characteristics, like adsorption, how dosage can alter wine flavour, and its efficacy based on pH levels, were required.

An apt response for carbon would reflect understanding its role in removing unwanted odours/flavours and decolourising wines. It would also explain how carbon works in the wine and how it can be beneficial and detrimental to its quality if misused.

A satisfactory response to dimethyl dicarbonate (DMDC) would include its application as a sterilising agent, recognising it as a protective measure against refermentation and spoilage yeasts. It would also contain the mode of DMDC action and important usage considerations such as correct dosage and setup costs.

A last thought is how to answer these types of questions. The question stipulated 'concise notes', yet some candidates answered the question in an essay format, presumably more time-consuming to write and more confusing for the examiner to mark. This question also highlighted an ongoing issue of students sharing similar examples across questions, regardless of the question being answered. Having an armament of real-life examples (rather than just educational seminars) is invaluable when writing the exam.

6. Are wild ferments worth the risk? Assess the advantages and disadvantages of choosing not to use cultured yeasts for fermentation.

The examiners expected a comprehensive analysis of the topic, including accurate definitions, an explanation of processes, and an assessment of the advantages and disadvantages associated with the choice of wild yeasts. Candidates were expected to support their points with relevant examples and conclude their report with an informed and reasoned viewpoint on the question asked.

This was a popular question and most candidates showed a basic understanding of wild fermentation, but few showed the appropriate depth of technical and practical knowledge of the realities and risks of fermenting with wild yeast. In particular, this was a question that cried out for understanding the main species of yeasts involved in wine fermentation. Many scripts failed to mention any yeast strains by name, and many of those that did couldn't spell *Saccharomyces cerevisiae*, let alone some of the wild yeast names.

Several candidates did not address the advantages and disadvantages aspect of the question. They seldom addressed the 'risk' aspect in detail, with very few candidates weighing up the economic and logistical risks associated with wild fermentation against the sensory rewards that wild fermentation can bring.

The better answers understood the sequential involvement of yeast types, with *Saccharomyces cerevisiae* primarily responsible for completing the fermentation. Spoilage microbes were poorly covered overall, with few candidates going into sufficient detail about the different organisms. Many scripts cited the contribution to the quality of a wild ferment through the lens of complexity but could have broken this down more into particular nuances and flavours. Lastly, introductions seldom defined the terms in sufficient detail, whilst conclusions rarely related to the question.

Breaking the answer down, one needs to grasp fermentation kinetics, what happens if wild yeasts take part and the main hazards of using them. Only a few scripts showed understanding of precisely how and why a slow fermentation could result in bacterial spoilage, and whilst *Brettanomyces* and *Saccharomyces* were frequently discussed, other microbes didn't get much attention. Some answers erroneously conflated wild fermentation with the production of natural wines. Others assumed incorrectly that issues with wild ferments were inevitable and could not be alleviated. Those winemakers choosing a path to wild ferment would undoubtedly take precautions to mitigate the potential problems of wild fermentation. These could include using turbid juice (or higher NTU juice), of which I saw no mention in the scripts I marked, as well as the use of 'pied de cuve' and rigorous monitoring of tanks and barrels to prevent a stuck ferment before it becomes terminal.

Several students used the cost argument, and whilst some justified this nicely, it would be a secondary factor for the winemaker given the cost savings are pretty minimal spread on a per litre basis, and more so given the inherent issues of wild ferment and further costs. A few answers made mention of reduced potential alcohol using a wild ferment, but very few were able to justify why. And in tandem, no one suggested that a commercial yeast company may also be looking into this area. Concerning natural and certified biodynamic wines, the better candidates recognised that lower limits of SO₂ increase the propensity for bacterial spoilage.

Examples often stated what a producer does rather than explaining why they choose to use/not use a particular practice and, therefore, did not add weight or substance to arguments. Yet, many students presented excellent and applicable examples, and example-rich essays often answered the two aspects of the question well

Paper three report 2023: Paper chair, Caroline Gilby MW

General comments

Paper 3 is all about ensuring that the hard work in producing wine (covered in papers 1 and 2) is not lost by poor handling or packaging, and it tests the understanding that goes with ensuring wine reaches the consumer in the expected condition. It is a technical paper and appropriate technical knowledge is expected, including actual numbers, measurements, and correct units. It also requires candidates to apply their knowledge to situations that might be encountered in the real world of wine handling. There is no substitute for visiting wineries, labs, and bottling lines to understand how processes and topics being tested work in practice, and this is a useful way of gathering material for use as examples too.

In 2023, strong candidates demonstrated appropriate technical knowledge, supported by real world evidence and examples. However, many others lacked factual knowledge, failed to adapt their answers to the specific question being asked, or ignored part of the question (e.g. failing to compare and contrast, or cover the “how” part of a question). Use of examples needs to be relevant – there is no point in talking about Penfolds, Roussillon, Mateus Rosé or Washington State for a question on New Zealand Sauvignon Blanc. Examples should be used to validate arguments.

Factual errors undermine confidence in everything a candidate has written. Statements like “good SO₂ management” or “low SO₂” or “lower oxygen” suggest the candidate does not know the required facts. Key knowledge for MW candidates includes: EU limits for SO₂ use and units (including organic wines) and clarity about whether referring to total or free SO₂; typical levels of free SO₂ for various styles; molecular SO₂ and pH; pore sizes for filtration; common wine additives and processing aids and what they do (legal limits and levels used in practice – there were several errors on this, such as potassium sorbate as anti-bacterial, CMC for protein haze, DMDC for tartrates and much more). It is worth double-checking these too – we saw molecular SO₂ stated at 0.8g/l and dissolved oxygen at 1.5 g/l. Inaccurate use of “sulphur” instead of sulphites or sulphur dioxide continues to crop up – elemental sulphur is a solid yellow mineral that is not added to wine in that form.

Failing to define key terms from the question was commonplace. It is much harder to write a clear argument without definitions and easy to miss part of the question. A good introduction can be brief and to the point, defining terms and setting out ground to be covered. One examiner commented, “In many scripts there was too much coverage on re-presenting the question and not enough on actually answering the question.” In one case as much as 40% of the content was spent on opening and closing paragraphs.

A few candidates wrote well but didn’t answer the question so could not score good marks. One examiner commented, “It was apparent students had had presentations or written essays on the topics covered generally by questions 2 and 3 in advance of the exam, but they seemingly lacked the ability/wider knowledge to adapt and apply that general knowledge to the specifics of the exam questions. They also tended to recount what they did know without examining this critically.”

Another point made by an examiner is that many essays are purely descriptive, focusing on setting out technical knowledge rather than on *applying* this knowledge to the question asked. A number of candidates submitted unfinished answers to the questions, suggesting that they didn't manage their time appropriately in the exam.

1. As a quality control (QC) manager for a large EU-based supermarket chain, you are auditing the QC and bottling procedures for one of your own-label suppliers for red and white wines. What would be your key areas of focus, and why?

This question was testing understanding of wine quality control analyses and treatments required for bottling to ensure a safe, stable quality product which meets EU regulations. The key point here is that the answer should have been written from the point of view of an auditor, not an onsite manager.

An audit report should have covered the following areas:

- Procedures, Policies and Certifications
- Quality management systems and certifications for bottling plant e.g. HACCP
- Suppliers should have documented validation procedures which assess ongoing safety, regulatory and quality criteria for all products supplied to supermarket.
- Product specification validation and verification, and checks on packaging specifications
- Full records of analyses and treatment decisions, sampling
- Procedures for breakages
- Traceability and lot code checks in place
- Staff training and policies in place for personnel
- Records for dry goods, glass, hard plastics.
- Environmental policy

This was an essay topic that required a considerable number of key points to be covered and many candidates did not cover enough ground, or drifted into unnecessary areas especially from grapes and through the winemaking process, or bulk shipping. Many answers simply described bottling procedures. Most essays missed some key parts. Several of the answers were written as if the candidate was talking the examiner through the bottling process (thus providing a general overview) without focus on the areas that would be audited by a QC manager. Some otherwise good answers were not supported by examples, which let them down from a final mark perspective. This was a technical question that revealed that many candidates lacked the necessary depth to answer to a level that would have achieved a passing mark.

2. Compare and contrast the technical implications of three non-glass packaging options available to a wine producer.

The most important function of food packaging is to protect and preserve the quality of the food or beverage stored within it. The primary aim of food packaging is to provide a barrier against oxygen, carbon dioxide, moisture, light, and still be able to preserve the flavour and aroma of the product. Another requirement of the package is that should be inert to flavour migration from the package to the product or adsorption of flavours from the product to the package. The current wine market has wine packaged in a wide range of materials including glass, a variety of plastics (it was important to

specify which plastic as they differ in their technical properties), aluminium, and plastic covered paper (Tetra Pak).

For this question, candidates who scored poorly either lacked sufficient technical knowledge or didn't read the question carefully enough so didn't note that a discussion of *technical* implications of using the various options was what was asked for. Market perceptions and customer acceptance were not required here.

3. Compare and contrast how stability is achieved in the following white wines:

- a. entry-level New Zealand Sauvignon Blanc;**
- b. domaine-bottled Meursault; and**
- c. Alsace Sélection de Grains Nobles Riesling.**

The question required candidates to look at similarities and differences in achieving stability for three different styles of white wine, so they needed to be clear about how the three styles are made. To achieve a pass, answers needed to go beyond merely describing how stabilisation is achieved for each.

Stability includes protein, cold, colour and microbial stability. Methods of achieving stability range from natural (essentially, time) to manipulated in order to get a wine stable to meet a bottling date or commercial requirements. Wine should not deteriorate in its intended shelf life. In summary, Entry-level NZ Sauvignon needs to be bottle-ready and on the market quickly, so requires processing to ensure the wine is stable; a domaine-bottled Meursault has time due to the wine style (fermented dry, full MLF, likely extended barrel ageing in cool cellar) so stability becomes more natural and therefore requires fewer winemaking inputs; a Sélection de Grains Nobles Riesling has the added complication of high laccase levels that require consideration, high residual sugar and possibly malic acid with risk of refermentation, along with the protein and cold stability issues. There may also be a problem of filterability.

It was clear that few candidates had detailed knowledge of processes - still less of the factors dictating these - used in relation to high-quality white Burgundy and even fewer had this knowledge in relation to SGN. Generally, the New Zealand Sauvignon Blanc was covered the best of the three styles, but this was not sufficient on its own to achieve a pass. Few candidates could draw on examples in relation to these styles and instead referred to producers making different styles in different regions, suggesting they hadn't grasped the role that examples should play in their answers. Other failings that were noted included little or no reference to colour stabilisation, and the failing of many scripts to cover the basics of bentonite and cold stabilisation. Many near-passes lacked depth of technical conviction and clarity of reasoning.

4. You are the head quality control (QC) manager for a major multiple retailer. Write a report detailing the implications of reducing post-fermentation sulphur dioxide additions in wine.

This was a technical question that required not just technical knowledge but also understanding of how the implications of reducing the level of sulphur dioxide were relevant for multiple wine styles. These two parts went hand in hand. Very few candidates demonstrated the ability to answer this question fully. Most of the answers gave a run down on the effects of low sulphur dioxide. The best

included a good range of wine styles in their report but also noted the key points around molecular sulphur and its importance/relevance to the question while noting EU legal parameters within which retailers must work.

It was relevant to cover the protective role of SO₂ additions in wine post-fermentation (anti-oxidant, anti-microbial, anti-enzymatic etc), as well as the importance of molecular free SO₂ and its relationship to pH (0.8mg/l molecular SO₂ and red wine at least 0.5mg/l molecular SO₂ to be considered safe from a change in state). Candidates should have been aware of the tendency for producers to reduce SO₂ use, including consumer sensitivity and the rise of consumer interest in “natural” wines, leading to pressure therefore to reduce SO₂ additions.

Risks of reducing SO₂ include oxidation and off-flavours, consumer dissatisfaction and returned product, with all the associated costs for large multiple retailer as well as the potential for reduced shelf-life of the product.

The question required candidates to be specific about SO₂ levels across various wine styles. The report should have covered why SO₂ is important, and the implications of a high or a low level at various points post-fermentation. For white and rosé wines, implications of reduced levels of SO₂ include oxidation, premature browning/development of colour, loss of aroma and flavour, refermentation, spontaneous malolactic, growth of bacteria, microbial spoilage, taints like mousiness, Plus risk of refermentation if residual sugar present. Reds may naturally be more stable thanks to tannins and polyphenols but also risk oxidation, browning, loss of aroma and flavour, refermentation, growth of bacteria and yeast especially *Brettanomyces*, thus microbial spoilage. Practical implications for wine handling might include: clean, botrytis-free fruit, scrupulous winery hygiene at all stages, minimising oxygen exposure during racking and handling, storing with no ullage or under inert gas, keeping temperatures at 10C or less, fermenting to dryness, 100% malolactic to remove substrates for refermentation, sterile filtration prior to bottling or bulk shipping, considering adjusting pH during winemaking for greater effectiveness of lowered SO₂ level, considering technological additives like chitosan derivatives, use of lysozyme, DMDC and addition of ascorbic acid to reduce SO₂ use.

Very few candidates provided the answer in a report format which also meant offering a recommendation or recommendations. One examiner summed up, “At times when reading a candidate’s answer, I tried to put myself in the shoes of a senior manager of a multiple retailer receiving this paper and asked myself does this ‘report’ outline the question at hand, provide a concise technical summary without being beyond the understanding of a colleague, highlight those wine styles with options to modify then summarise what would be required from a winery and the risks associated to the retailer and finally a recommendation?”

Paper four report 2023: Paper chair, Simon Thorpe MW

1. Which sectors of the wine industry are currently delivering the best return on investment?

The question asks about a very specific element of the financial analysis of business: return on investment (ROI). For those candidates who did not have a clear understanding of what this means, it was very difficult to answer the question well.

ROI is a simple ratio that divides the net profit (or loss) from an investment by its cost. Because it is expressed as a percentage, you can compare the effectiveness or profitability of different investment choices.

This question was asked primarily because the wine industry is typically regarded as having very low returns on investment. There are often low returns because of the high levels of investment required to establish wine businesses (throughout the supply chain from vineyards to retail outlets), high costs of doing business, long term investments, and then relatively low profit margins.

Many of the poor answers simply focused on profit and described those businesses or sectors which are profitable. This did not answer the question as some of these have very low ROI. Good answers clearly defined ROI and highlighted examples of areas such as luxury brand (production or ownership) distribution where there are significant barriers to entry (and therefore high margins can be maintained), and producers optimising the Direct to Consumer channel. It was also valid to talk about business models with low investment models, virtual wineries for example, or those companies who can generate a profit quickly such as fine wine brokers or even consultants.

2. Which sectors of the wine industry are doing the most to advance the sustainability agenda?

The question is straightforward and looks to test candidates' knowledge of the whole of sustainability, not just those aspects concerning the environment. A great many candidates focussed solely on the environment which was problematic and caused quite a few to fail.

A good answer needed a clear definition and there have been many versions over the years. However, it's a clear mistake to concentrate only on the environment when the general understanding includes environment, social and financial motivations. Nevertheless, there were plenty of candidates who tackled this well and those with the greatest success included analysis of those making the biggest positive impact, referencing the question 'driving the agenda'. This concentrates primarily on those producers leading the way (with lots of good examples of associations and individual wineries doing great work in this area) and those further down the supply chain such as Systembolaget or retailers across the globe. It was necessary to provide some structure rather than just write a list of those showing good practice, just to make a cogent and readable answer.

3. You are the global brand director for an internationally established, prestige Champagne house. You have an annual marketing budget of five million euros (approximately five million U.S. dollars), not including price promotions, and have been asked to present a three-year brand plan. What are your key areas of focus and expenditure?

This was a question which required some setting of parameters in order to formulate a cogent argument – there is obviously no formula for a 'correct' answer – and there was more than one way to write a successful answer. Therefore, it was up to the candidate to define the positioning of their brand, what its priorities and market proposition might be and therefore what its strategy should focus on. Good answers gave clear parameters at the start and then made logical suggestions as to the strategy.

Often there was simply an analysis of the champagne industry, which was fine as long as it was used to inform the strategy for the brand described in the question. But this did not really address what we were looking for in the question.

The best answers gave a clear brand positioning at the start and then assessed the objectives and challenges for their brand, whether it be a mature brand looking to grow into new markets, a brand looking to premiumise, a brand looking to attract a younger consumer base, or one looking to build some new product development into their portfolio. There were lots of options and it was an opportunity for candidates to show creativity if answered correctly.

One note is that the levels of spend did need to be realistic – meaning the appropriate amount to be spent out of the \$5m on the various executions. Often the numbers suggested were way too high or too low.

4. What are the biggest challenges facing wine brand managers looking to create and maintain a loyal customer base in today's competitive market?

The question asks about the challenges of today's market in relation to customer loyalty. These are the two key elements to the question and any discussion beyond this was largely irrelevant.

Brand loyalty is a particularly difficult element in the wine industry with so many competitors at all sectors and a consumer with so much choice, information and desire for experimentation. One good way of answering this question was to dissect the market into segments and then address the loyalty challenges in each, using good and differentiated examples from multiple markets. Building loyalty for mass-market, mainstream wine brands is very different from luxury brands; the consumer behaves differently and requires different treatment.

Once a good structure was established, it was relatively straightforward to think of the challenges faced, whether it be constant social media bombardment, aggressive pricing or challenges of supply and demand.

One element it would have been good to see more analysis on is around the direct to consumer and 'buy local' consumer trends. Increasingly, brands both large and small are looking to engage directly with consumers – a look at this area would have helped build an excellent answer for sure.

5. What are the key elements of a successful relationship between a wine producer and their distributor?

This was a popular and generally well-answered question. The relationship between producer and distributor is fundamental to how our industry works and therefore it is perhaps not surprising that this was well answered. A lot of people work in this part of the industry!

In order to stand out here it was important to have great examples and some good data. It's easy to point to the requirement for both parties to have a clear and shared objective, but that needs evidencing by perhaps an example of good and bad practice in this area.

Equally, the question doesn't provide an obvious structure and there would be a temptation to rush into this and just list those elements which come to mind. It was preferable to segment the answer, whether that be by route to market, by destination market, by prestige/premium/mainstream market positioning. All of these approaches would have been suitable.

Good answers gave a range of different examples across the different supply chain models and had clear structure. Less successful candidates just discussed the theory at a superficial level without demonstrating a clear knowledge of how relationships really work.

6. Outline the challenges and opportunities facing wine producers in California. What is the long-term outlook for this region?

This was not a popular question and not especially well answered by those who did attempt it. It was obviously important to have a good understanding of the dynamics of the Californian industry before deciding to answer this question and it was clear that some who attempted to answer were not close enough to be able to go into depth with good examples. The questions required an assessment of challenges and opportunities so this provided some ready-made structure. Some people answered using the PESTLE model of analysis and this works really quite well when assessing a region. There are elements in each of the political, economic, social, technological, legal and environmental arenas which are relevant here. Using this model also ensured candidates didn't focus too much on one single element, which was a downfall for many who focused excessively on climate change and its consequences.

Good answers were able to point to production and commercial challenges ranging from mainstream through to super-luxury wines. Very few pointed to how Californian wine fits into the context of the global industry and the challenges and opportunities this brings, whether that be production challenges in Europe or political challenges for some countries in China. Again, examples were key, supported with accurate data. Without these it was hard to produce a successful answer.

Paper five report 2023: Paper chair, Ken MacKay MW

General Comments

The candidates who did well on paper 5 this year have absorbed the advice from previous examiners' reports. Although there is not a completely fixed assemblage for a great answer, there are certain essentials in the blend, which come to mind after reviewing this year's exam. Sticking with the wine analogy:

Avoid confusing labels. Clear, precise and sufficiently detailed definitions are essential at the start, so that your subsequent arguments are absolutely unambiguous and easy for the examiners to understand.

Aim for a balanced blend & avoid simple monovarietals. Weaker answers often fall down because they communicate one clear view, but are unbalanced. Whatever your own informed opinions might be, never fail to highlight the valid arguments for opposing views.

Complexity usually impresses more than simplicity. Broad answers, covering as many points as you can, to more fully explore and answer the question, will serve you better than going into too much depth on fewer points.

Include the geeky tech specs. Always provide specific, named and (where at all possible) quantified evidence to support your arguments: scientific study findings, market research figures, consumption figures, legal limits, relative approximate costs of different closures and containers for wine, and so on. Gathering these headline facts and figures should be a key part of your exam preparation strategy. For example, in question 2 this year, many failed to clarify the specific alcohol levels cited in the OIV definition of 'wine'.

Put your product in great packaging & market yourself well. Presentation of your arguments is important. This year there were quite a few 'stream of consciousness' answers, cramming several unrelated points into extremely long, unfocused paragraphs. While there is no set prescription for success shorter, highly focussed, point-specific paragraphs can be more impactful and effective.

Avoid obvious faults. There were some howlers this year in the spelling of wine appellations, brands and prominent individuals' names. Whilst the MW exam is not a spelling test, and very reasonable latitude is given if English is not your first language, there is no real excuse for errors of spelling in wine-specific names or terms. If you are going to cite a wine-related example, you need to know how it's spelled: it's your own choice of example, after all.

Produce something that reflects an individual terroir. Try to be really original with your examples, if you can, as it helps you reflect greater personal insight. Almost all candidates answering question 1 this year mentioned Jancis Robinson's documenting of bottle weights in her reviews, although there are so many other examples of individuals and businesses in the trade who have championed lightweighting.

Judge the importance of everything you put in your blend. Many papers were beautifully written, but lacked the necessary critical analysis to show greater understanding, independence of thought and overall mastery of the topic.

Remember your bottom-line: always think about the commercial and practical realities. Paper 5 questions might be thought-provoking, but they are not meant to be abstract and answers should always be clearly grounded in reality. Try to take a step back from every question and consider: what might the impacts of all this be on profitability, personnel, logistics and pricing?

1. What is the future of traditional wine packaging?

What seemed to be a relatively straightforward question proved to be the most challenging on this year's paper, with the lowest pass rate. An initial definition of 'traditional wine packaging' was needed and this was expected to be the longest-established packaging choice commonly used for wine today: glass bottles, paper labels, natural cork, foil/laminate capsules and cardboard/wooden cases. Answers needed a critical analysis of traditional packaging and the challenges and potential of innovative options, supported by evidence, in the form of specific figures and global examples. Pass level answers needed to take both an industry and consumer perspective, across various concerns, such as: image/marketing; convenience; sustainability; cost; fashion; quality; shelf-life etc., and to identify who or what was driving change (or resistance to it).

Many poorer answers made very superficial reviews, with many candidates inappropriately deciding to limit their discussion to the container, discussing alternatives to glass bottles, without considering the other key elements of the package (such as the label, closure, foil/capsule and case/carton). Poorer answers also often failed to answer clearly the question as to what 'the future' was. Passing papers did more than merely provide the general landscape of packaging, they supported this with relevant facts, figures, and statistics to strengthen their argument (relative costs, recyclability, national recycling rates, carbon footprints, OTRs etc.) Better answers considered: legal/trade/environmental and logistical issues; the volume of the container, as well as the material; refillables; national and industry commitments; and contrasted different global producer, trade (on/off), supply chain and (multiple) consumer perspectives. The best papers considered solutions for the future; other industries for guidance; shifts in consumer behaviour; technical considerations for marketing and authenticity/security; as well as the ways in which recent global events have impacted the supply chain.

2. Is alcohol an essential component of high-quality wine?

There is no right or wrong answer to this question, but candidates needed to explore different perspectives (rather than simply express one strong opinion), and make a balanced, realistic argument, with plenty of global examples, and reach a very clear conclusion, to pass. A definition of 'high-quality' and 'wine' was essential in the introduction, and the latter had to show an awareness of legal definitions, linked to specific ABV thresholds. Answers needed to display an appreciation that views on what constitutes 'high-quality' are not simply black and white, but complex, diverse and producer, category, market- and consumer-sector specific. While the examiners were not looking for a purely paper 2-style essay, a sound grasp of wine's production and chemical composition was needed. Alcohol has so many varied roles in wine – as a solvent and aroma vehicle, as a contributor

to texture and body, as a preservative and antimicrobial, as an intoxicant, and even as a basis for taxation – that there was no shortage of topics to consider when delivering a full, rounded paper 5 answer. Finally, it was essential that answers cover several styles such as late harvest wines, ice wine, rosé, sparkling wines, and fortified wines, in addition to still whites and reds.

Surprisingly, such a fundamental topic for our industry yielded many short, superficial answers. Seeing “high-quality” in the question led many candidates to adopt a “BLIC(A)” structure, which often restricted answers to too limited a seam of content. While the good essays revelled in the question's breadth; poorer ones approached the subject with a narrow focus - for instance, dwelling purely on alcohol's role in vinification, or overlooking it entirely. Poorer answers frequently interpreted the question as a debate about no/low alcohol wine and how good/bad it is. Although this could legitimately form part of an argument, it needed to be balanced with other points. There were many ways to elevate and enrich answers: examples include consideration of whether fortified and sparkling wines are a special case, evaluating the validity of various means of dealcoholisation, comparing wine to other alcoholic drinks categories, and weighing up social/health issues against aesthetic factors. The best candidates combined at least some of these topics with a solid central analysis of alcohol's contribution to quality, enhanced with specific evidence, real-world examples and appropriate rhetorical flair. Strangely, remarkably few papers pointed out that alcohol is intoxicating and attempted to relate how this might influence perceptions and experience.

3. In your opinion, what should be on a wine label, and why?

This was one of the better answered questions on this year's paper. Answers needed to examine the diverse roles of 'front' and 'back' labels, from multiple supply chain perspectives, offering specific, global examples of what 'should' be on a label (and why), linking this to who benefits (or is disadvantaged): regulatory bodies, regional trade associations, producers, importers, wholesalers, retailers, sommeliers, consumers (of different kinds, in different markets) etc. A good answer needed to consider various 'needs' for a wine label, including: legal; industry and consumer-protection: product safety and traceability, allergens and authenticity/quality designations; branding and marketing; informative (abv, volume, style, food pairing, kosher status); ethical (fair trade, carbon footprint, recycling, environmental credentials, vegan); health advisory (alcohol units, enjoy in moderation messaging, calorific content, residual sugar levels). As this was an 'in your opinion' question, papers ultimately needed a clear, well-argued, critical conclusion on what 'should' be on a wine label.

Poorer papers missed important elements - particularly: bottler/importer traceability and lot codes (although the latter may obviously also appear elsewhere on the bottle); legally-required or marketing indications of taste or style; 'field of view' requirements and distinctions between front and back labels - and often failed to offer clear personal opinions or specific examples. Better answers considered authenticity elements for fraud prevention; the redesign and reprint cost to producers of regular label changes; and displayed global awareness of current legislation and anticipated changes such as the new EU regulations. Many mentioned QR codes to good effect, but few questioned how many global consumers typically interact with these elements on product packaging. The best answers explored different demographic preferences and linked their arguments into how elements, such as ethical or environmental credentials, improved listing success and consumer purchase rates; or the role of own-labels or 'tertiary brands'. Overall, very few candidates engaged with the prospect

of 'tobacco style' packaging and whether it was important for the industry to try to prevent progression towards this; or mentioned relevant consumer research on buying decision hierarchies – i.e., what people look for most on the label; or critically reviewed the appropriateness of (variable) tolerances for ABV on the label in different markets.

4. 'Minerality is one of the most overused yet misunderstood words in the wine industry.'

Discuss

'Minerality' in the world of wine is variously used to refer to rock minerals (inorganic compounds), nutrient minerals (soluble ions), or as a sensory descriptor for aromas and flavours redolent of the earth, soil and rocks (e.g., flint, chalk); but it is also perhaps often confused with other characteristics and the concept of 'terroir'. Minerality as a term bridges the fault-line between a scientific approach to wine and a more artistic and culturally-informed approach. Exploring both perspectives was essential in answering this question, as was providing a broad-perspective attempt to define what minerality should mean, in wine industry terms, and how it should be used. It was also important to consider different consumer perspectives and critically assess the relevance of the term for marketing (labels, wine lists, websites etc.) and wine journalism. Examples supporting and opposing its use, for different reasons, were essential, as was clarity as to why minerality might be 'overused' and 'misunderstood'.

The question required a degree of confidence around technical aspects relating to minerality, which is perhaps why relatively few candidates attempted it. Poor answers generally fell down due to a lack of evidence of knowledge of technical aspects, relevant to supporting (and disputing) possible sources of perceptions of mineral flavours/textures in the vineyard, winery and maturation process, and failed to take a balanced view (for example, not recognising the positive impact and association the descriptor might have on a wine's perceived reputation or value). Better answers were aware of the relatively recent appearance of the term in the wine lexicon; addressed aspects of minerality in relation to soil chemistry and vine absorption, as well as potential viticultural and winemaking influences; and also explored the trade/consumer/media usage of the term. Good answers backed up opinions with wide-ranging examples, relating to scientific as well as artistic aspects of the term, and also had the confidence not just to discuss whether the term was misused/misunderstood, but also to express definite opinions of what would be appropriate, for communication with consumers as well as professionals.

5. Is a wine's ability to age important?

It was essential, in answering this question, to clarify that 'ability to age' goes beyond preservation of quality (stability or perishability), to encompasses a perceived ability to improve in quality. Wine is virtually unique in that it can continue to evolve and improve, sometimes for decades, in its final packaging after the point of purchase. Answers needed to explore how the importance of ageability varies enormously for individual wines and styles, and by perspective (for example: producers, retailers, critics, journalists, trade members; different target markets and consumers), and to look at the question from multiple angles. As well as the aesthetic effect of certain wines tasting better with bottle age; there is the complex socio-cultural side: if wine were perishable, there would be no vertical tastings, no lasting record of vintage and winemaking influences, no chance to witness a particular wine developing over time and to share and discuss that experience within a community of critics or connoisseurs. Furthermore, there is a substantial economy and labour market sustained by

wine's ageability – brokers and auction departments, storage facilities and cellar builders, fine wine critics and authenticators, and so on – which needed to be weighed in any answer. And who can ignore the emotional component of receiving a gift of wine from one's birthyear, or drinking an aged bottle from a historically significant bygone age? Finally, answers absolutely needed commercial realism: as the vast majority of wine is consumed shortly after purchase, *a particular* wine's ability to age is, most often, of little importance.

Many poorer answers conflated ability to age with mere stability, rather than also taking it to mean wine's ability to improve in bottle; reflected misunderstandings concerning the fine wine market or the chemistry of bottle ageing; were very simplistic and lacked relevant examples; failed to mention that regional classifications had specific ageing requirements; or did not properly weigh the significant commercial importance of ageing fine wines for investment and its support for ancillary industries. Good answers avoided those pitfalls, while dealing comprehensively with the topic across a range of styles (encompassing more than just still light wines, but also touching on sweet, fortified and sparkling styles) - identifying that the ability to age well is something that remains central to the appeal, romance, value and cultural appreciation of wine for many - and highlighted the commercial realities at play, regarding cash flow and storage practicalities. The most sophisticated approaches introduced points of greater insight, such as distinguishing the importance of rarity and perceived versus actual ability to age; or observing that the emotional effect is heightened by the coincidental fact that fine wine ages on a roughly human timescale; and also highlighted evolving historical perspectives on the question, perhaps linking this to changing lifestyles, with reduced home ownership and disposable income, combined with more of an 'instant gratification' ethic, making wine ageing less feasible or attractive.